

**CS1010 Advanced Placement Credit
Examination Preparation for
S&T2025 Scholars (Set 9)**

CS1010 Placement Test (Self-Diagnostic Assessment Question)

<https://www.comp.nus.edu.sg/programmes/ug/exemptions/sda-programmeth/>

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1. A game of "Pick Up Sticks" is played between two players with an initial pile of sticks. Each player takes turns in picking out 1 to 3 sticks from the pile. Whoever takes the last stick loses the game.

Suppose it is your turn to pick sticks from the pile, which of the following situation would be least favourable to you?

A pile of 7 sticks

A pile of 8 sticks

A pile of 6 sticks

A pile of 10 sticks

A pile of 9 sticks

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2. A bunny would like to get from one side of the river to the other side. There are n rocks separating the two sides with equal spacing, and the bunny can only hop across j number of rocks (or steps) in a single jump.

Using an example of $n = 4$ and $j = 3$, suppose the bunny needs to get from s to d with rocks denoted by x in between. Let capital letters denote the current position of the bunny.

Initially:

S x x x x d

Hop #5 (hoping 3 steps towards right):

s x x x x D

Hop #1 (hopping 3 steps towards the right):

s x x X x d

Which of the following configurations of n rocks and j steps would result in the bunny **NOT** being able to reach its destination? You may assume that n is an even number, and $j \leq n$.

Hop #2 (hopping 3 steps exceeds d and bounces back):

s x x x X d

The least common multiple of $(n + 1)$ and j is odd.

Hop #3 (hopping 3 steps towards the left):

s X x x x d

The least common multiple of $(n + 1)$ and j is even.

Hop #4 (hopping 3 steps exceeds s and bounces back):

s x X x x d

The greatest common divisor of $(n + 1)$ and j is odd.

The greatest common divisor of $(n + 1)$ and j is even.

None of the above.

3. Santa is about to deliver the 2000 wrapped boxes of identical pairs of socks to 2000 families. However, Santa's elves have mistakenly wrapped a pair of heavily soiled socks. Time is running out. Which one of the 2000 boxes contains the dirty socks? There is no time to unwrap all the boxes, find the one with the dirty socks, and wrap the rest up again. We know that the soiled socks are heavier. There is a huge balancing scale which the elves may use, but there isn't enough time to put the boxes one at a time on the balancing scale. How can you help Santa to locate the dirty socks as quickly as possible? Using your method, at most how many times the balancing scale will be used to locate the box containing the dirty socks?

Divide the boxes by a quarter, that is, 500 boxes in each group, to locate the group that contains the dirty socks. This group is then divided by a quarter again, and the process is repeated until the box that contains the dirty socks is located. This will take at most 9 weighings.

Divide the boxes into groups of hundreds. At most 19 weighings are needed to locate the group with the dirty socks. Then at most another 99 weighings to locate the box with the dirty socks among the hundred boxes.

Divide the boxes by half, that is, 1000 boxes in each group, to locate the group that contains the dirty socks. This group is then divided by half again, and the process is repeated until the box that contains the dirty socks is located. This will take at most 9 weighings.

Divide the boxes into groups of ten. At most 199 weighings are needed to locate the group with the dirty socks. Then at most another 9 weighings to locate the box with the dirty socks among the ten boxes.

Divide the boxes by half, that is, 1000 boxes in each group, to locate the group that contains the dirty socks. This group is then divided by half again, and the process is repeated until the box that contains the dirty socks is located. This will take at most 11 weighings.

4. Suppose the following cards are shuffled and put in a pile, face down:



Assuming that "A", "J", "Q" and "K" have values 1, 11, 12 and 13 respectively, find out the order of the cards in the pile from top to bottom if you are given the following clues:

1. The eight is somewhere above the "J".
2. A two is directly above the five.
3. The spade is somewhere above the heart.
4. One black card is directly above the other black card.
5. The bottom card's value is the sum of the values of three other cards.
6. A red card is on the top of the pile.

From top to bottom: 8 diamond, 2 diamond, 5 club, Jack spade, 2 heart, Queen diamond.

From top to bottom: Queen diamond, Jack spade, 5 club, 8 diamond, 2 diamond, 2 heart.

From top to bottom: 8 diamond, 2 heart, Jack spade, Queen diamond, 5 club, 2 diamond.

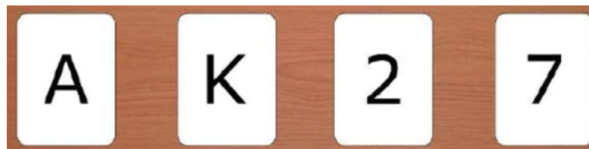
From top to bottom: 2 diamond, 5 club, Jack spade, 2 heart, Queen diamond, 8 diamond.

From top to bottom: Jack spade, 2 heart, 2 diamond, 5 club, 8 diamond, Queen diamond.

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5. Suppose you are told that if a card has a vowel on one side, it must have an even number on the other side. Given the following four cards, you are to choose two cards to turn over to prove or disprove the above rule. Which two cards will you choose?



Cards "K" and "7".

Cards "A" and "7".

Cards "A" and "K".

Cards "K" and "2".

Cards "A" and "2".

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- J1** I think it must assume that behind a letter, it is a number. Or flipping cards "A" and "K" and flipping cards "A" and "7" are both correct.

Jessica, 7/20/2025

C1. Given the following code fragment:

```
int list[5] = {3, 2, 4, 1, 0}, i = 0, temp;

while (i < 5) {
    if (list[i] != i) {
        temp = list[i];
        list[i] = list[list[i]];
        list[list[i]] = temp;
    } else {
        i++;
    }
}
```

{4, 3, 2, 1, 0}

{0, 1, 2, 3, 4}

{1, 2, 4, 3, 0}

{0, 0, 0, 0, 0}

An infinite loop occurs.

What are the contents of array list after executing the program?

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C2. Given the following structure definition:

```
typedef struct {
    char name[40];
    int age;
} Person;
```

Which of the following functions correctly return(s) the sum of ages in the first n elements of array pa?

i.

```
int ageSum(Person pa[], int n) {
    int sum = 0, i;

    for (i = 0; i < n; i++) {
        sum += pa[i].age;
    }
    return sum;
}
```

i. only.

i. and ii. only.

i. and iii. only.

ii. and iii. only.

All of i., ii. and iii.

ii.

```
int ageSum(Person pa[], int n) {
    Person *p = pa;
    int sum = 0, i;

    for (i = 0; i < n; i++) {
        sum += (p+i)->age;
    }
    return sum;
}
```

iii.

```
int ageSum(Person pa[], int n) {
    int sum = 0, i;

    for (i = n-1; i > 0; i--) {
        sum += pa[i].age;
    }
    return sum;
}
```

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C3. Given the following function:

```
void foo(int *x, int *y, int *z) {
    int temp;

    temp = *x;
    *x = *y;
    *y = temp;

    *z = *x + *y;
    return;
}
```

What is the output of the following code fragment?

```
int i = 1, j = 2, k = 3;

foo(&i, &j, &k);
printf("%d %d %d", i, j, k);
foo(&i, &i, &i);
printf("%d %d %d\n", i, j, k);
```

1 2 3; 1 2 3

2 1 3; 2 2 3

2 1 3; 4 1 3

2 1 3; 4 4 3

None of the above.

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C4. Study the following doIt function.

```
void doIt(int x, int y, int *count) {
    int result;

    if ((x == 1) || (y == 1)) {
        (*count)++;
    } else {
        doIt(x-1, y, count);
        doIt(x, y-1, count);
    }

    return;
}
```

What is the output of the following program fragment?

```
int count = 0;
doIt(2, 3, &count);
printf("%d\n", count);
```

2

3

5

10

None of the above.

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C5. You are tasked to decode a message.

Each letter in the original message has been shifted four places to the left in the alphabet.

So to decode, each letter has to be shifted four places right with wrap-around, a should be e, z should be d, and so on.

As an example, the coded message:

zk ukq hkra yo1010?

should be decoded to become:

do you love cs1010?

You are given the following decode function.

```
void decode(char *str) {
    while (*str != '\0') {
        if (*str >= 'a' && *str <= 'z') {
            // code fragment
        }
        str++;
    }
}
```

Assuming that the message is written using lower case letters, which of the following code fragments perform the decoding on string str?

i.

```
*str += 4;
if (*str > 'z') {
    *str -= 26;
}
```

i. only.

ii.

```
*str -= 22;
if (*str < 'a') {
    *str += 26;
}
```

i. and ii. only.

i and iii. only.

ii. and iii. only.

iii.

```
*str = (*str - 'a' + 4)%26 + 'a';
```

All i., ii. and iii.